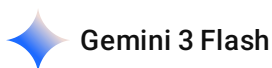


So for a homogenous elastic system with $E=70$ GPa and $\nu=0.3$ that is stretched by 8% in the x directions, calculate the σ_{xx} and σ_{yy} in these three conditions:

1-fully plain strain

2-plain strain in z direction, but $\sigma_{yy}=0$

3-fully plain stress



python

```
import numpy as np

E = 70e9 # Pa
nu = 0.3
eps_xx = 0.08

# Case 1: Fully Plane Strain (epsilon_yy = 0, epsilon_zz = 0)
# sigma_xx = [E / ((1+nu)*(1-2nu))] * (1-nu) * eps_xx
# sigma_yy = [E / ((1+nu)*(1-2nu))] * nu * eps_xx
factor1 = E / ((1 + nu) * (1 - 2 * nu))
sigma_xx_1 = factor1 * (1 - nu) * eps_xx
sigma_yy_1 = factor1 * nu * eps_xx

# Case 2: Plane strain in z (epsilon_zz = 0), but sigma_yy = 0
# epsilon_zz = 0 => sigma_zz = nu * (sigma_xx + sigma_yy) = nu * sigma_xx
# epsilon_xx = (1/E) * (sigma_xx - nu*sigma_zz) = (1/E) * (sigma_xx - nu^2 * sigma_xx) =
# (sigma_xx / E) * (1 - nu^2)
sigma_xx_2 = (E * eps_xx) / (1 - nu**2)
sigma_yy_2 = 0.0

# Case 3: Fully Plane Stress (sigma_zz = 0)
# Usually implies sigma_yy = 0 for a uniaxial test in x
sigma_xx_3 = E * eps_xx
sigma_yy_3 = 0.0

print(f"Case 1: sigma_xx = {sigma_xx_1/1e9:.2f} GPa, sigma_yy = {sigma_yy_1/1e9:.2f} GPa")
print(f"Case 2: sigma_xx = {sigma_xx_2/1e9:.2f} GPa, sigma_yy = {sigma_yy_2/1e9:.2f} GPa")
```

```
print(f"Case 3: sigma_xx = {sigma_xx_3/1e9:.2f} GPa, sigma_yy = {sigma_yy_3/1e9:.2f} GPa")
```

text

Case 1: sigma_xx = 7.54 GPa, sigma_yy = 3.23 GPa

Case 2: sigma_xx = 6.15 GPa, sigma_yy = 0.00 GPa

Case 3: sigma_xx = 5.60 GPa, sigma_yy = 0.00 GPa

To calculate the stresses σ_{xx} and σ_{yy} for the given homogeneous elastic system, we use the generalized Hooke's Law.

Given Parameters:

- Young's Modulus (E): 70 GPa
- Poisson's Ratio (ν): 0.3
- Applied Strain (ϵ_{xx}): 0.08 (8%)

1. Fully Plane Strain

In a "fully plane strain" condition (often called uniaxial strain), all lateral strains are constrained to zero.

- **Conditions:** $\epsilon_{yy} = 0$ and $\epsilon_{zz} = 0$.
- **Formula for σ_{xx} :** $\sigma_{xx} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\epsilon_{xx}$
- **Formula for σ_{yy} :** $\sigma_{yy} = \frac{E\nu}{(1+\nu)(1-2\nu)}\epsilon_{xx}$

Calculation: $\sigma_{xx} = \frac{70 \times (1-0.3)}{(1+0.3)(1-2 \times 0.3)} \times 0.08 = \frac{49}{0.52} \times 0.08 \approx 7.538 \text{ GPa}$ $\sigma_{yy} = \frac{70 \times 0.3}{0.52} \times 0.08 \approx 3.231 \text{ GPa}$

Results:

- $\sigma_{xx} = 7.54 \text{ GPa}$
- $\sigma_{yy} = 3.23 \text{ GPa}$

2. Plane Strain in z direction, but $\sigma_{yy} = 0$

This is a mixed condition where the z -direction is constrained (plane strain), but the y -direction is free to expand or contract.

- **Conditions:** $\epsilon_{zz} = 0$ and $\sigma_{yy} = 0$.

- **Relation:** For $\epsilon_{zz} = 0$, we have $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$. Since $\sigma_{yy} = 0$, then $\sigma_{zz} = \nu\sigma_{xx}$.
- **Formula for σ_{xx} :** Substituting into $\epsilon_{xx} = \frac{1}{E}[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$, we get: $\epsilon_{xx} = \frac{1}{E}[\sigma_{xx} - \nu(0 + \nu\sigma_{xx})] = \frac{\sigma_{xx}}{E}(1 - \nu^2) \implies \sigma_{xx} = \frac{E\epsilon_{xx}}{1 - \nu^2}$

Calculation: $\sigma_{xx} = \frac{70 \times 0.08}{1 - 0.3^2} = \frac{5.6}{0.91} \approx 6.154 \text{ GPa}$

Results:

- $\sigma_{xx} = 6.15 \text{ GPa}$
- $\sigma_{yy} = 0 \text{ GPa}$

3. Fully Plane Stress

In a "fully plane stress" condition (standard uniaxial stress), all lateral stresses are zero. The material is free to deform in both y and z directions.

- **Conditions:** $\sigma_{yy} = 0$ and $\sigma_{zz} = 0$.
- **Formula for σ_{xx} :** $\sigma_{xx} = E\epsilon_{xx}$

Calculation: $\sigma_{xx} = 70 \times 0.08 = 5.6 \text{ GPa}$

Results:

- $\sigma_{xx} = 5.60 \text{ GPa}$
- $\sigma_{yy} = 0 \text{ GPa}$

Summary Table

Condition	σ_{xx} (GPa)	σ_{yy} (GPa)
1. Fully Plane Strain	7.54	3.23
2. Plane Strain (z), $\sigma_{yy} = 0$	6.15	0.00
3. Fully Plane Stress	5.60	0.00